Utility-Invariant Support Selection and Eventwise Decoupling for Simultaneous Independent Multi-Outcome Bets

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Abstract

For simultaneous independent events with finitely many outcomes, consider the expected-utility problem with nonnegative wagers and an endogenous cash position. We prove a short support theorem for a broad class of strictly increasing strictly concave utilities. On any fixed support family and at any optimal portfolio with positive cash, summing the active first-order conditions and comparing that sum with cash stationarity yields the exact identity

$$\frac{\lambda}{K_\ell^{(U)}} = \frac{1 - P_{\ell,A}}{1 - Q_{\ell,A}},$$

where $P_{\ell,A}$ and $Q_{\ell,A}$ are the active probability and price masses of event ℓ, λ is the budget multiplier, and $K_\ell^{(U)}$ is the continuation factor seen by inactive outcomes of that event. Consequently, after sorting each event by the edge ratio $p_{\ell i}/\pi_{\ell i}$, the exact active support is the eventwise union of the single-event supports, and this support is independent of the utility function. The single-event utility-invariant support theorem is already explicit in the free-exposure pari-mutuel setting in Smoczynski and Miles; the point of the present note is that the simultaneous independent-events analogue follows from the same state-price geometry once the right continuation factor is identified.

1 Introduction

For a single multi-outcome event, the active-set structure of the full Kelly problem is classical; see, among others, Kelly [1], Rosner [3], Smoczynski and Tomkins [5], and Whelan [6]. In state-price language, cash is an implicit all-state position, so active wagers merely top up favorable outcomes beyond the cash floor; see Long [2]. For general increasing concave utility in the free-exposure pari-mutuel setting, Smoczynski and Miles [4] prove that the optimal set of horses is independent of utility.

The purpose of this note is to prove the simultaneous independent-events analogue in a short self-contained form. The decisive structural observation is that, for a fixed event, every inactive outcome sees the same background wealth from the rest of the portfolio. This common continuation factor is the multi-event analogue of the single-event cash floor. Once that object is named, the proof is a short stationarity-decomposition argument: sum the active first-order conditions, condition the cash stationarity equation on one event, and subtract.

The result isolates a clean division of labor. Utility curvature affects the *active weights*, but not the *support*. In the simultaneous independent problem, support selection decouples eventwise and is governed by exactly the same threshold $(1 - P)/(1 - Q)$ as in the single-event problem. This sharpens the structural interpretation of simultaneous Kelly-type optimization beyond the algorithmic literature on many simultaneous bets; see Whitrow [7].

2 Model and admissible utilities

Consider $m \geq 1$ independent events. Event ℓ has outcomes $i \in \{1, \ldots, n_\ell\}$, subjective probabilities

$$p_{\ell i} > 0, \qquad \sum_{i=1}^{n_\ell} p_{\ell i} = 1,$$

and state prices

$$\pi_{\ell i} > 0.$$

A portfolio consists of a cash position $c \geq 0$ and nonnegative wager sizes $g_{\ell i} \geq 0$, subject to the bankroll constraint

$$c + \sum_{\ell=1}^{m} \sum_{i=1}^{n_\ell} \pi_{\ell i} g_{\ell i} = 1. \tag{1}$$

If the realized product-state is $X = (X_1, \ldots, X_m)$, terminal wealth is

$$W(X) = c + \sum_{\ell=1}^{m} g_{\ell, X_\ell}. \tag{2}$$

We restrict attention to portfolios satisfying

$$W(x) > 0 \qquad \text{for every product-state } x.$$

Definition 1. *An* admissible utility *is a function $U : (0, \infty) \to \mathbb{R}$ such that:*

(U1) *$U \in C^1((0, \infty))$,*

(U2) *$U'(w) > 0$ for every $w > 0$,*

(U3) *U is strictly concave on $(0, \infty)$.*

The optimization problem is

$$\max \ \Phi(c, g) := \mathbb{E}\big[U(W(X))\big] \quad \text{subject to (1), } c \geq 0, \ g_{\ell i} \geq 0, \ W(x) > 0 \ \forall x. \tag{3}$$

For each outcome define the edge ratio

$$r_{\ell i} := \frac{p_{\ell i}}{\pi_{\ell i}}. \tag{4}$$

A support family is a tuple $A = (A_1, \ldots, A_m)$ with $A_\ell \subseteq \{1, \ldots, n_\ell\}$. We say that (c, g) is *supported on A* if

$$g_{\ell i} > 0 \text{ for } i \in A_\ell, \qquad g_{\ell i} = 0 \text{ for } i \notin A_\ell.$$

For fixed ℓ, define the background wealth from all other events by

$$R_\ell(X_{-\ell}) := c + \sum_{r \neq \ell} g_{r, X_r}. \tag{5}$$

Then on the slice $\{X_\ell = i\}$ one has

$$W(X) = g_{\ell i} + R_\ell(X_{-\ell}).$$

In particular, if $i \notin A_\ell$ then $g_{\ell i} = 0$ and the slice wealth is exactly $R_\ell(X_{-\ell})$, independent of which inactive outcome is being tested.

3 Fixed-support stationarity and reduced costs

The next proposition records the only fixed-support facts needed for the theorem.

Proposition 2 (Fixed-support first-order and reduced-cost conditions)**.** *Fix a support family $A = (A_1, \dots, A_m)$, and let (c, g) be an optimal portfolio for the restriction of (3) to portfolios supported on A. Assume $c > 0$. Then there exists a multiplier $\lambda > 0$ such that*

$$\lambda = \mathbb{E}\big[U'(W(X))\big], \tag{6}$$

and, for every event ℓ and every active outcome $i \in A_\ell$,

$$p_{\ell i}\, \mathbb{E}_{-\ell}\big[U'(g_{\ell i} + R_\ell(X_{-\ell}))\big] = \lambda \pi_{\ell i}. \tag{7}$$

If $j \notin A_\ell$, define

$$K_\ell^{(U)} := \mathbb{E}_{-\ell}\big[U'(R_\ell(X_{-\ell}))\big]. \tag{8}$$

Then the one-sided directional derivative in the feasible direction $g_{\ell j} \mapsto g_{\ell j} + \varepsilon$ at $\varepsilon = 0^+$ is

$$p_{\ell j} K_\ell^{(U)} - \lambda \pi_{\ell j},$$

and optimality therefore forces the reduced-cost inequality

$$p_{\ell j} K_\ell^{(U)} \le \lambda \pi_{\ell j}. \tag{9}$$

In particular, for fixed ℓ, every inactive outcome shares the same continuation factor $K_\ell^{(U)}$.

Proof. Restricting to support family A means that the variables $g_{\ell i}$ with $i \notin A_\ell$ are fixed at zero, while c and the active variables $g_{\ell i}$ with $i \in A_\ell$ vary subject to (1). Because U is strictly concave and wealth is affine in the decision variables, the restricted objective is strictly concave. Standard Kuhn-Tucker conditions therefore apply.

Write the Lagrangian

$$L(c, g; \lambda) = \Phi(c, g) - \lambda \left(c + \sum_{\ell, i} \pi_{\ell i} g_{\ell i} - 1 \right).$$

Because $c > 0$, complementary slackness yields stationarity with respect to c. Since $\partial W(X)/\partial c = 1$ in every state,

$$\frac{\partial \Phi}{\partial c} = \mathbb{E}\big[U'(W(X))\big],$$

which gives (6).

If $i \in A_\ell$, only the slice $\{X_\ell = i\}$ contributes to the derivative with respect to $g_{\ell i}$, so

$$\frac{\partial \Phi}{\partial g_{\ell i}} = p_{\ell i}\, \mathbb{E}_{-\ell}\big[U'(g_{\ell i} + R_\ell(X_{-\ell}))\big].$$

Stationarity in the active variable $g_{\ell i}$ yields (7).

If $j \notin A_\ell$, then $g_{\ell j} = 0$. The directional derivative of the objective in the feasible direction $g_{\ell j} \mapsto g_{\ell j} + \varepsilon$ at $\varepsilon = 0^+$ equals

$$p_{\ell j}\, \mathbb{E}_{-\ell}\big[U'(R_\ell(X_{-\ell}))\big] = p_{\ell j} K_\ell^{(U)}.$$

The corresponding directional derivative of the Lagrangian is therefore $p_{\ell j} K_\ell^{(U)} - \lambda \pi_{\ell j}$, and optimality implies (9). The common-factor claim follows directly from (8). □

4 The threshold identity

For a support family A, define the active probability and price masses of event ℓ by

$$P_{\ell,A} := \sum_{i \in A_\ell} p_{\ell i}, \qquad Q_{\ell,A} := \sum_{i \in A_\ell} \pi_{\ell i}. \tag{10}$$

If the outcomes of event ℓ are sorted so that

$$r_{\ell 1} \geq r_{\ell 2} \geq \cdots \geq r_{\ell n_\ell},$$

and $A_\ell = \{1, \ldots, k_\ell\}$ is a prefix, we abbreviate

$$P_{\ell,k_\ell} := \sum_{i \leq k_\ell} p_{\ell i}, \qquad Q_{\ell,k_\ell} := \sum_{i \leq k_\ell} \pi_{\ell i}.$$

Theorem 3 (Utility-invariant threshold identity)**.** *Fix a support family $A = (A_1, \ldots, A_m)$, and let (c, g) be optimal for the restriction of (3) to portfolios supported on A. Assume $c > 0$. Then for every event ℓ,*

$$\frac{\lambda}{K_\ell^{(U)}} = \frac{1 - P_{\ell,A}}{1 - Q_{\ell,A}}, \tag{11}$$

where λ is the multiplier from Proposition 2 and $K_\ell^{(U)}$ is defined by (8).

Proof. Fix an event ℓ. Summing the active stationarity equations (7) over $i \in A_\ell$ gives

$$\lambda Q_{\ell,A} = \sum_{i \in A_\ell} p_{\ell i}\, \mathbb{E}_{-\ell}\big[U'(g_{\ell i} + R_\ell(X_{-\ell}))\big]. \tag{12}$$

On the other hand, conditioning the cash stationarity equation (6) on X_ℓ yields

$$\begin{aligned} \lambda &= \sum_{i \in A_\ell} p_{\ell i}\, \mathbb{E}_{-\ell}\big[U'(g_{\ell i} + R_\ell(X_{-\ell}))\big] + \sum_{j \notin A_\ell} p_{\ell j}\, \mathbb{E}_{-\ell}\big[U'(R_\ell(X_{-\ell}))\big] \\ &= \sum_{i \in A_\ell} p_{\ell i}\, \mathbb{E}_{-\ell}\big[U'(g_{\ell i} + R_\ell(X_{-\ell}))\big] + (1 - P_{\ell,A}) K_\ell^{(U)}. \end{aligned} \tag{13}$$

Substituting (12) into (13) gives

$$\lambda = \lambda Q_{\ell,A} + (1 - P_{\ell,A}) K_\ell^{(U)},$$

which rearranges to

$$\lambda (1 - Q_{\ell,A}) = (1 - P_{\ell,A}) K_\ell^{(U)}.$$

Dividing by $K_\ell^{(U)} > 0$ proves (11). □

Corollary 4 (Eventwise prefix support and exact decoupling)**.** *Assume, in addition, that each event is sorted so that*

$$r_{\ell 1} \geq r_{\ell 2} \geq \cdots \geq r_{\ell n_\ell}.$$

Let (c, g) solve the simultaneous problem (3) with $c > 0$. Then for each event ℓ there exists an integer $k_\ell \in \{0, 1, \ldots, n_\ell\}$ such that

$$g_{\ell i} > 0 \text{ for } i \leq k_\ell, \qquad g_{\ell i} = 0 \text{ for } i > k_\ell.$$

Moreover,

$$r_{\ell,k_\ell+1} \le \frac{1-P_{\ell,k_\ell}}{1-Q_{\ell,k_\ell}} \qquad (k_\ell < n_\ell), \tag{14}$$

and the exact support of event ℓ is the same support obtained by solving the corresponding single-event problem with the same probabilities and prices. Hence the simultaneous exact support is the eventwise union of the single-event supports.

Proof. Let $A_\ell := \{i : g_{\ell i} > 0\}$. If $i \in A_\ell$, then $g_{\ell i} > 0$, so strict concavity implies

$$U'(g_{\ell i} + R_\ell(X_{-\ell})) < U'(R_\ell(X_{-\ell}))$$

for every $X_{-\ell}$. Taking expectations and using (7) and (8),

$$\frac{\lambda \pi_{\ell i}}{p_{\ell i}} = \mathbb{E}_{-\ell}[U'(g_{\ell i} + R_\ell(X_{-\ell}))] < K_\ell^{(U)}.$$

Thus every active outcome satisfies

$$r_{\ell i} > \frac{\lambda}{K_\ell^{(U)}}. \tag{15}$$

For every inactive outcome $j \notin A_\ell$, the reduced-cost inequality (9) gives

$$r_{\ell j} \le \frac{\lambda}{K_\ell^{(U)}}. \tag{16}$$

Hence, after sorting by decreasing $r_{\ell i}$, the active set of event ℓ must be a prefix $\{1, \dots, k_\ell\}$. Applying Theorem 3 to that prefix support yields

$$\frac{\lambda}{K_\ell^{(U)}} = \frac{1-P_{\ell,k_\ell}}{1-Q_{\ell,k_\ell}},$$

so (16) for the next inactive outcome becomes exactly (14). This is the same stopping rule as in the single-event problem, so support discovery decouples eventwise. □

Corollary 5 (Single-event utility-invariant support)**.** *For $m = 1$, the optimal active support is the greedy prefix determined by*

$$r_{k+1} \le \frac{1-P_k}{1-Q_k},$$

independently of the admissible utility U.

Proof. When $m = 1$, one has $R_1 \equiv c$ and therefore $K_1^{(U)} = U'(c)$. Theorem 3 and Corollary 4 then yield the stated cutoff immediately. □

Remark 6 (One-dimensional reduction in the single-event problem)**.** *Once the single-event support A is known, the active first-order conditions imply*

$$U'(W_i) = \frac{\lambda}{r_i} \qquad (i \in A),$$

so

$$W_i = (U')^{-1}\left(\frac{\lambda}{r_i}\right) \qquad (i \in A).$$

$$\theta_A := \frac{1 - P_A}{1 - Q_A},$$

Theorem 3 gives $\lambda/U'(c) = \theta_A$*, hence*

$$c = (U')^{-1}\left(\lambda \frac{1 - Q_A}{1 - P_A}\right).$$

The budget constraint becomes a single scalar equation for λ*:*

$$(1 - Q_A)(U')^{-1}\left(\lambda \frac{1 - Q_A}{1 - P_A}\right) + \sum_{i \in A} \pi_i (U')^{-1}\left(\frac{\lambda}{r_i}\right) = 1.$$

For CRRA utility $U(w) = w^{1-\gamma}/(1-\gamma)$*, this equation is explicit.*

Remark 7 (Log utility and CRRA as special cases)**.** *For* $U(w) = \log w$*, Theorem 3 recovers the support threshold in Long [2]. For CRRA utility* $U(w) = w^{1-\gamma}/(1-\gamma)$ *with* $\gamma > 0$*, the theorem shows that the active support is unchanged across the entire CRRA family: risk aversion changes the active weights, but not the active set.*

Remark 8 (Boundary case $c = 0$)**.** *When* $c = 0$*, cash stationarity becomes one-sided. More precisely, there is a slack term*

$$\nu := \lambda - \mathbb{E}[U'(W(X))] \geq 0.$$

Repeating the proof of Theorem 3 gives

$$\lambda(1 - Q_{\ell,A}) = (1 - P_{\ell,A})K_\ell^{(U)} + \nu,$$

so

$$\frac{\lambda}{K_\ell^{(U)}} = \frac{1 - P_{\ell,A}}{1 - Q_{\ell,A}} + \frac{\nu}{(1 - Q_{\ell,A})K_\ell^{(U)}}.$$

Thus the exact utility-invariant threshold identity is an interior phenomenon. This clarifies why exogenous exposure constraints or fractional-betting constraints generally destroy support invariance.

Remark 9 (Independence as the structural boundary)**.** *The proof uses independence at exactly one point: for an inactive outcome* $j \notin A_\ell$*, the continuation factor*

$$K_\ell^{(U)} = \mathbb{E}_{-\ell}[U'(R_\ell(X_{-\ell}))]$$

must be the same for every such j*. Under dependence, the conditional law of* $X_{-\ell}$ *given* $X_\ell = j$ *varies with* j*, so different inactive outcomes see different continuation factors. The conditioning identity (13) then no longer collapses to a single scalar threshold. Independence is therefore not a mere convenience; it is the exact structural boundary of the theorem.*

Remark 10 (Strict vigorish and canonical uniqueness)**.** *Suppose every event has strict overround,*

$$\sum_{i=1}^{n_\ell} \pi_{\ell i} > 1 \qquad \text{for every } \ell.$$

Then the fair-event cash-shift degeneracy is absent: one cannot move a constant amount from cash into every outcome of a single event without changing the budget. If, in addition, there are no exact threshold ties, then the eventwise prefix support in Corollary 4 is unique. If equality occurs at a threshold, the natural canonical tie-break is to prefer cash, i.e. to treat equality outcomes as inactive.

References

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